

Editorial

*“Improvements in information technology are expanding the feasible regions of the design space for organizations.”
[Tom Malone, “Free On The Range”]*

*“It is Enterprise which builds and improves the world’s possessions... If Enterprise is afoot, Wealth accumulates whatever may be happening to Thrift; and if Enterprise is asleep, Wealth decays, whatever Thrift may be doing.”
[J M Keynes, “Treatise on Money”]*

*“There were times when it seemed to him that different parts of him were not all under the same management”
[Russell Hoban, “The Lion of Boach-Jachin and Jachin-Boaz”]*

The world in which the modern enterprise operates is radically changing. Increasing deregulation in the telecommunications sector, coupled with the convergence of media, computing and communications industries, is resulting in a highly competitive environment. Acquisitions and mergers predominate in the telecommunications sector as a means of entering new market spaces. This is symptomatic of both convergence in the industry and small agile start-up companies being a major source of innovation. To survive, successful enterprises have to be structured to be able to respond to change. Agility, i.e. responsiveness and flexibility in the face of evolving markets, is key to business success.

For an organisation to be agile it requires its underlying information systems to be built with agility in mind. In BT’s various joint ventures, disparate enterprise information systems have to be integrated to provide a transparent, seamless service to customers. This represents a significant engineering challenge. Such systems have to be designed not just to meet existing requirements but with a view to their potential future uses.

This issue

This edition brings together strands of research which together provide an insight into how the large modern enterprise can be made agile. The variety of technical subjects touched upon emphasises the multi-disciplinary nature of this topic.

The papers demonstrate how emerging technologies can support an intelligent self-organising enterprise infrastructure which enables the:

- co-ordination of business activities and collaboration of teams within and between enterprises,
- information and knowledge management across enterprises.

Technologies are explored which emphasise the ‘bottom-up’, decentralised, non-hierarchical management of enterprises as opposed to the traditional rigid centralised hierarchies originally espoused by Max Weber in 1940s. The former offers the advantages of adaptability and flexibility necessary in today’s economy, while the latter provides stability and rigidity which inhibits change. As an aside, it is interesting how technological infrastructure increasingly reflects changes in organisational structures and theory.

The technologies described in this issue will help BT and its customers develop, deliver and maintain products and services with the speed, flexibility, quality and economy that is necessary in the highly competitive telecommunications market of today. The paper by Jones et al (p 15) introduces these topics by exploring the application of lean engineering principles to telecommunications services.

The remaining papers fall broadly into four themes — modelling enterprises, managing processes, managing information and knowledge, and decentralisation and emergence — as detailed below.

Modelling enterprises

It is often said that one cannot manage what one cannot measure, but it is also true that one cannot manage (or indeed measure) what one cannot model. Enterprise modelling is necessary both to understand the existing constituents and properties of an enterprise, and to decide how to change it in order to improve its properties. The main property of interest here is agility, the ability to respond to changing market demands and opportunities. The paper by Koubarakis and Plexousakis (p 23) outlines a useful theoretical framework for enterprise modelling. This is mathematically formal which makes it amenable to analysis so that the properties of an enterprise may be characterised. Agility remains difficult to define, let alone measure, but at the very least establishing and maintaining a model enables one to keep track of the enterprise as it changes and record the reasons.

An enterprise is a complex network of entities and relationships that can be studied from many different viewpoints. One important relationship is that of an enterprise (or its constituent units) with its customers and suppliers. Russell et al (p 36) use a concept termed supply stamps to characterise these relationships, pointing out that they depend on the position of the unit in the supply chain. Transformation stamps are a related concept describing the relationship between a unit's input and output, and the way one is transformed into the other.

There is a great deal in common between enterprise modelling and requirements analysis for the business process software that supports it. The history of software engineering has seen design and programming languages becoming ever less oriented towards computer architecture and ever more oriented towards the way in which the analyst or programmer views the requirements. Since the mid 1980s the object metaphor has become dominant, and UML is emerging as the *de facto* standard design notation. More recently, other high level abstractions have been introduced, and UML is being extended to include them. These include agents and roles, both of which also feature in enterprise modelling. The paper by Kendall (p 46) uses role modelling to describe patterns of relationships among software agents, which in many ways mirror the relationships found in human organisations.

Managing processes

The next grouping of papers takes a process-oriented view of the enterprise. A business process comprises a number of activities which collectively realise some market need or business objective. In this issue, five papers explore the theme of business process management with a view to increasing the flexibility of processes and their management. Hutton's paper (p 58) introduces workflow technology and how it is currently being employed in BT. Workflow separates the business logic embedded in a business process from the underlying information systems which implement the process activities. This paper reflects the current state-of-the-art technologies available for process management. Shepherdson et al (p 65) take this technology one step further by describing how workflow technology can be significantly enhanced through intelligent agent technology. This automates routine activities currently undertaken by managers such as negotiation for resources, and exception handling. Mehandjiev and Odgers (p 72) introduce the softer human aspect of these technologies. They emphasise the importance of making such technology human centred and promote the use of technology to empower users. This theme is continued in Swaby et al (p 78) who look at business activities more as a co-operative working environment. They demonstrate how virtual reality interfaces can support work co-ordination. Lastly, 'ASOPE and the fabled promise of process knowledge capture and reuse' (p 91) by Thompson et al looks further into the future where an organisation's business processes are bespoke, tailored for the individual needs of all those stakeholders involved in the particular process, such as the customer, engineers, etc.

Managing information and knowledge

In the knowledge economy, information is a valuable asset if it can be exploited. Unfortunately, system heterogeneity, poor data quality and a lack of semantics leave many corporate information systems underused and underexploited. Equally, the dynamic creation of enterprises through partnerships or mergers places additional demands on information systems when they are required to interoperate. One key problem with exchanging information between organisations is that they use different information models with different

associated semantics (i.e. meanings). This is often true with different parts of a single organisation. The semantics are not explicitly represented and invariably remain tacit in the minds of users and developers. Achieving interoperability at a semantic level is often referred to as the ontology problem. The papers in this section explore how emerging technologies can address this problem.

Cui et al (p 98) introduce the section with a description of on-going research into developing a one-stop knowledge shop for BT's information systems. Their approach aims to make the semantics of corporate information explicit via an ontology which can be used to support *ad hoc* information requests. In the Internet, the consumer is forced to discriminate between a bewildering array of content — information overload is commonplace. The ability to personalise or tailor content to individual consumers' needs is increasingly important. Krohn et al (p 108) outline recent research into modelling information-seeking behaviour in the Internet and how this can be used to tailor information to specific user groups.

Fiddian et al (p 117) report on project KRAFT which developed a prototype system for the dynamic integration of diverse information sources in support of the network design process. They use software agents to mediate between users' information requests and heterogeneous vendor data bases cataloguing network components. Lastly, Kim et al (p 131) apply the TOVE ontology to building quality systems, and demonstrate how it can support quality problem identification and tracing in quality control applications.

Decentralisation and emergence

The debates between proponents of central control and local autonomy are fierce and bloody (consider, for example, the American Civil War, or parliamentary debates on Maastricht or the Euro). Each faction claims (with some justification) that the god of agility is on its side. The centralists argue that a single point of control with a global view enables a decision to be made taking all factors into account, an optimised and co-ordinated plan formulated, and instructions propagated so that each unit in the organisation knows how to play its part. The enterprise changes direction with the grace and precision of the Coldstream Guards trooping the colour. The libertarians respond that in practice this leads to plans that are rigid, unrealistic, obsolete before they are issued, and that it is best to delegate decisions to personnel (or software) closely associated with the sources of information and means of taking action. The danger here is that these autonomous units will independently make conflicting decisions.

Thomas Malone (Professor at the MIT Sloan School of Management and founder and director of the MIT Center for Co-ordination Science) has provided an excerpt from a previously published paper that helps introduce the decentralisation and emergence theme (p 141). This is reproduced with kind permission from the Sloan Management Review. He argues that modern communications and information technology makes a middle way advantageous. This he describes as a connected decentralised approach, though other papers in this issue use simply decentralised. In this approach, local decisions are made autonomously, but the decision-makers are mutually aware so that decisions made by one influence those made by others.

Kearney and Merlat (p 145) commence their paper with a similar line of argument applied primarily to the organisation of distributed software systems. Specifically they advocate a view of large-scale software systems as autonomous adaptive agents buying and selling services from each other. The organisation of the system as a whole is then emergent, i.e. it arises as a result of repeated cycles of adaptation and economic interaction at the agent level. The paper by Smith et al (p 157) then extends this approach by adding evolutionary computing mechanisms inspired by the work of Holland, acting over longer time-scales than the economic ones. Emergent phenomena in complex systems can be undesirable as well as beneficial, and further research is required to understand how to achieve the latter while suppressing the former.

The final paper by Merlat (p 168) shows how mobile agents can be used to implement a decentralised constraint satisfaction algorithm. The application domain used to illustrate this is eCommerce — specifically travel management.

What lies ahead?

On the threshold of the new millenium BT is looking forward to a business environment rich in challenges and opportunities. Software technology is a course of many of these challenges — it is changing the nature of our core business. It is also lowering the threshold for new start-up companies entering the fray. By

their very nature, these new entrants are agile:

- their small size means that strategic decisions can be taken and implemented rapidly,
- they are unencumbered by an existing product base and the management, process and technology base required to support it and so can optimise themselves for their target product area,
- they are able to utilise state-of-the-art infrastructure.

BT must both acquire the advantages possessed by the new entrants and harness its own strengths — breadth and depth of experience, synergy through horizontal and vertical integration, existing infrastructure, customer base, brand, etc.

The papers in this issue point out ways that software infrastructure and related work can assist in achieving this goal. Although they have been presented as distinct, they are in fact all interrelated. The overview of the enterprise modelling theme described the enterprise as a complex network viewable from many perspectives, and the different sections of this issue are perspectives rather than disjoint disciplines. An enterprise modeller sees collections of information as components of the enterprise, yet from an information management perspective, the enterprise model is information to be managed. Furthermore the techniques used in ontologies and enterprise models are very similar — in fact the term enterprise ontology is often used. Similarly the enterprise model both defines the business processes and is a resource required to enact the process in a flexible way.

The papers on emergence recognise that the enterprise is not a static thing, nor is it a passive entity that is occasionally reconfigured by an external management figure. Management is part of the enterprise, and the processes that modify it are business processes like those that deal with supplying services to customers. To be agile, BT's processes must sense trends in the trading environment and cause corresponding adaptive changes within the organisation. Thus we move to a view of an enterprise as an organism rather than a mechanism.

One thing can be certain, despite all attempts at prediction, the future will surprise us — we must expect the unexpected and be prepared to take advantage of it. The work described in this issue will enable us to both shape the future and respond to it. In the words of Sir Karl Popper:

“We may become the makers of our fate when we have ceased to pose as its prophets.”

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